David Ricardo Coria Hernandez

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Education

University of Kansas, Lawrence, KS
PhD in Physics, **Advisor**: Ian Crossfield — July 2025
"Galactic Chemical Evolution & Giant Exoplanet Formation: Insights from Volatile Abundances"

Kansas State University, Manhattan, KS
Cum Laude, Bachelor of Science in Mathematics — May 2020
Cum Laude, Bachelor of Science in Physics — May 2020

Research Positions

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Postdoctoral Fellow — September 2025 – Present
Exoplanets @ UNM, University of New Mexico, **Advisor**: Diana Dragomir

Graduate Research Assistant — August 2020 – July 2025
KU ExoLab, University of Kansas, **Advisor**: Ian Crossfield

Research Summary

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I use high-resolution (Optical, NIR, MIR) spectroscopy to study the chemical composition of stars and measure individual elemental and isotopic abundances. I am particularly interested in how dwarf star abundances can be used to test galactic chemical evolution models and how they correlate to other stellar properties. I also measure volatile and refractory abundances in exoplanet host stars to predict the formation history of their companion exoplanets targeted by JWST and state-of-art ground-based observatories.

Awards and Honors

Alliance of Hispanic Serving Research Universities (HSRU) Postdoctoral Fellow	Present
KU "I Am First Too" Commemorative Poster Feature	Fall 2024
IGEN Travel Grant (An NSF INCLUDES Alliance) Recipient	Summer 2024
NASA SCoPE Seed Grant Recipient	Spring 2024
Barbara J. Anthony-Twarog Outreach Award	Spring 2024
NASA Exoplanet Explorers Program: Inaugural Cohort Member	2021
University of Kansas Graduate Fellow	Fall 2020 – Spring 2021
Hagan Scholarship Foundation Recipient	Fall 2016 – Spring 2020
McNair Scholars Program: Research Assistant	Fall 2018 – Spring 2020
Developing Scholars Program: Research Assistant	Fall 2016 – Spring 2019
Kansas State University Honor Roll	Fall 2016 – Spring 2020
Kansas State University Putnam (Distinguished University) Scholar	Fall 2016 – Spring 2020

Mentoring, Teaching, Tutoring

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- **REU Student: Sally Shepard** Summer 2026

I mentored undergraduate student Sally Shepard from Mount Holyoke College, MA. During this 10-week summer research experience, Sally learned the basics of stellar spectra analysis (flattening, RV correction, normalization, stacking, etc.), prepared a research poster, and presented the poster in an REU Symposium. Sally worked with data from high-priority JWST host star targets to explore the possibility of measuring CNO and rare P+S chemical abundances at different wavelength regimes (optical, Y band, H band).

Guest Lectures

- UNM ASTR 2110: Earth's Atmosphere Spring 2026
- UNM ASTR 2110: Earth's Interior Spring 2026
- KU ASTR 391: Radiation Spring 2023
- KU ASTR 191: Galaxy Classification/Evolution Spring 2023

- **Graduate Teaching Assistant: Contemporary Astronomy Lab** Fall 2023

KU ASTR 196 is an introduction to astronomical observations and modern data analysis methods. This is a descriptive rather than mathematical astronomy course focused on qualitative sky observations paired with computational data analysis. In this semester-long lab, students maintain an observing journal where they learn to identify solar system planets, moons and constellations and track how they move over time. I also guide students through several hands-on activities designed to give students a better understanding of how astronomers study planetary orbits, stellar classification, galaxy classification, and cosmic distances.

- **Instructor: Lawrence High School** Spring 2023

I served as the instructor for a research-based astronomy course at Lawrence High School, a minority-serving institution, where I guided students through the full scientific research process using real astronomical data. The course focused on analyzing data from the Virgo Filament Survey, teaching students how to manipulate large datasets using tools like Excel and TopCat. Students developed essential research skills, including conducting literature reviews, formulating research questions, processing and analyzing data, and interpreting their findings. Emphasizing science communication, I mentored students in designing professional research posters, which they presented at a student-led research symposium. This experience provided hands-on exposure to real astronomical research and fostered critical thinking, problem-solving, and data literacy skills. By creating an inclusive learning environment, I aimed to empower students from underrepresented backgrounds to engage in STEM and build confidence in scientific inquiry.

- **Private Tutor: TRIO, Upward Bound, McNair Scholars Program** 2021-2023

I tutored several undergraduates from academic success programs like TRIO, Veterans Upward Bound, and the McNair Scholars Program. I covered material from both math & physics courses including College Algebra, Trigonometry, Calculus I, II, and III, Physics I and II.

Service/Outreach

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Commented [DC5R4]: Review Panels
Commented [DC6R4]: Journal Review

Journal Referee/Review Panels

- Gemini LLP Review Panelist S2026
- NSF Review Panelist, Astronomy & Astrophysics Section S2026
- AJ Journal Referee S2024
- NASA XRP Executive Secretary S2021

Outreach

- **The Albuquerque Astronomical Society**: Member, Planetarium & Star Party Presenter Fall 2025 - Present
- **Astronomy Outreach Coordinator, KU Physics & Astronomy** Spring 2025

As the Astronomy Outreach Coordinator, I lead the department's astronomy public engagement, event organization, and hands-on educational experiences. I am responsible for planning and executing public telescope nights, planetarium shows reaching audiences ranging from K-12 students to the general public.

- **Traveling KU Planetarium: Co-Organizer and Presenter** 2021 - 2025

I organize and present planetarium shows to campus and other local groups with the Traveling KU Planetarium (a Digitarium Portable Planetarium System).

- **KU Telescope Nights: Co-Organizer** 2021 – 2025

I plan and run monthly observing nights on campus for students and the community: telescope observing of the moon, solar system planets, star tours, constellation identification, etc.

- **Pluto Day Celebration: Co-Lead Organizer** February 2025

I co-organized a celebration of the 95th Anniversary of Pluto's discovery, held at the Lawrence Public Library. We hosted lots of arts & crafts, a scale solar system demo, a liquid nitrogen demo, and a build-a-Pluto demo. I also designed Pluto-themed CubeWorlds using maps from Hubble and New Horizons.

- **Rocks & Rockets: KU Co-Organizer (Colby, KS)** August 2023

Presented planetarium shows for a public science event in one of the most rural places in the US!

Educational Resource Development S2026

- Finding Exoplanets with NASA's Universe of Learning September 2025

The Finding Exoplanets Program Guide provides informal educators with science background information, activity ideas, and other resources to engage audiences with NASA exoplanet science. The accompanying activities– "Exoplanet Detectives: A Transit Method Activity" & "Lights, Coronagraph, Action! An Exoplanet Direct Imaging Demo" – teach participants about planets outside our solar system and how astronomers find them.

- CubiMundos: Spanish CubeWorlds Present

A series of crafty, cube-shaped planets and moons. Now available en Español!

- NASA SCoPE Project w/ NASA UoL & GSAWN 2024-2025

Las Estrellas y sus Compañeros: Developing Bilingual Activities & Resources for NASA's Universe of Learning

This project seeks to develop new, engaging ways to share mainstream astronomical discoveries with local Hispanic communities. We build on the Girls STEAM Ahead with NASA (GSAWN) guidebook for educators by developing a new exoplanet-themed section with engaging activities and resources teaching the basics of exoplanet discovery and characterization. The project will also provide a professional Spanish translation of the entire GSAWN guidebook to boost the reach of the resource– not just to Spanish-speakers in Kansas, but across the nation and world.

Conference Organization

- **Cool Stars 22: Splinter Session Co-Organizer (San Diego. CA)** Summer 2024

Planet-Host Cool Dwarfs: Tracing Planetary Formation & Composition with the Star-Planet Connection

In this session, we cover current methods and future improvements for measuring the physical parameters and individual elemental abundances of cool dwarfs. We also discuss how the chemistry of these host stars relates to protoplanetary disk composition and subsequent planetary formation and evolution.

- **KU Physics & Astronomy Locally Organized Assembly Co-Organizer** 2023-2024

"PALOOZA" is an annual, GSO-sponsored research symposium open to both graduate and undergraduate students within the Geology, and Geography & Atmospheric Science departments to showcase our diverse research in a low-stakes and supportive environment.

- **NASA ExoExplorers: DEI Special Session at AAS** January 2023

A session to dissect how members of marginalized communities navigate unique challenges throughout their scientific careers and what the astronomy community can do to lessen this impact on early career scientists.

- **Advocacy for Kansas-based TRIO/McNair programs** Spring 2022

Wrote letters of support for TRIO staff advocating for continued federal funding of these programs to legislators in Washington, D.C.

Observing Experience

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Keck I & II 10m Telescopes, W. M. Keck Observatory, HI
- 2 nights on NIRSPEC; 2 nights HIRES

4m Anglo-Australian Telescope, Siding Spring Observatory, NSW, Australia
- 3 nights on Veloce

3m NASA Infrared Telescope Facility, HI
- 2 nights on iSHELL

8.1m Gemini South Observatory, Cerro Pachon, Chile
- **PI**: "Elemental Abundances of the Coolest HWO Targets"
- **Co-I**: "Exo-EASY: Elemental Abundances via SpectroscopY of Exoplanet-hosting Cool Dwarfs" (LP)
- **Co-I**: "Elemental Abundances of JWST's Exoplanet Host Stars" (Q)
- **Co**-I: "Mass and Spin-Orbit Alignment of a Temperate Neptune" (Q)

Conference and Public Presentations

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Conference Talks

- **Contributed Talk: HORS3S (Granada, Spain)** July 2026

Tracing Elemental and Isotopic CO Abundances in JWST Planet-Hosting K Dwarf Stars Using High-Resolution NIR Spectroscopy

- **Contributed Talk: Cool Stars 23 Splinter Session (Tokyo, Japan)** June 2026

Tracing Elemental and Isotopic CO Abundances in JWST Planet-Hosting K Dwarf Stars

- **Dissertation Talk: AAS 245 (National Harbor, MD)** January 2025

Galactic Chemical Evolution & Giant Exoplanet Formation: Insights from Volatile Abundances

- **Contributed Talk: Mid-America Regional Astrophysics Conference** December 2024
- **Contributed Talk: Cool Stars 22 Splinter Session (San Diego, CA)** June 2024

Exoplanetary Origins: Unraveling Planetary Formation and Accretion Histories with CNO Isotopologues; doi: 10.5281/zenodo.13619959

- **Contributed Talk: KU CREATORS Symposium** February 2024

Cosmic Close-Ups: The Past, Present, and Future of Direct Imaging in Exoplanets

- **Contributed Talk: AAS 243 (New Orleans, LA)** January 2024

Tracing Giant Exoplanet Formation Using Complementary Host Star CNO Abundances

- **Contributed Talk: Mid-America Regional Astrophysics Conference** November 2023

Tracing Giant Exoplanet Formation Using Complementary Host Star CNO Abundances

- **Contributed Talk: Towards Other Earths III (Porto, Portugal)** July 2023

CNO Isotope Ratios Across Exoplanet Systems: Implications for Planet Formation and Atmospheric Composition

- **Special Session: AAS 241 (Seattle, WA)** January 2023

It's Giving... Back: Advocating for Minority-Oriented Academic Success Programs as an Alum

- **Contributed Talk: IR 2022 (Remote)** February 2022

The Missing Link: Connecting Exoplanets and Galactic Chemical Evolution via Stellar Abundances: Isotopic Carbon and Oxygen Abundances in Solar Twin Stars

- **Contributed Talk: Exoplanet Explorers Science Series (Remote)** April 2021

The Missing Link: Connecting Exoplanets & Galactic Chemical Evolution via Stellar Abundances

Conference Posters

- **iPoster: AAS 241 (Seattle, WA)** January 2023

The Missing Link: Testing Galactic Chemical Evolution Models with the First Multi-Isotopic Abundances in Solar Twin Stars

- **iPoster: Cool Stars 21 (Toulouse, France)** July 2022

The Missing Link: Testing Galactic Chemical Evolution Models with the First Multi-Isotopic Abundances in Solar Twin Stars

- **Contributed Poster: Cool Stars 20.5 (Remote)** February 2021

Measuring CO Isotopic Abundance Ratios in Solar Twin Stars; doi: 10.5281/zenodo.4563216

Other Talks

- TAAS General Meeting — August 2026
- TAAS Portable Planetarium Show: The North Star and a Solar System Tour — 2025-2026
- AstroCoffee: Institute for Astronomy (University of Hawai'i at Mānoa) — April 2024
- **Planetarium Show: Rocks & Rockets (Colby, KS)** — August 2023

Intro to Celestial Navigation

- **Alumnx Panel: McNair Heartland Research Conference (KC, KS)** — 2021
- **Contributed Talk: McNair Heartland Research Conference (KC, KS)** — 2019

Simplified echellogram data reduction and spectral extraction via the Veloce Quick Look App

First Author Projects & Publications

1. **David R. Coria**, Martha Irene Saladino, Alexander Cotnoir, Amelia Chapman, Rutuparna Das, Varoujan Gorjian, Kimberly Arcand, Emma Marcucci, Denise Smith; *Program Facilitator Guide: Finding Exoplanets*, September 2025, NASA's Universe of Learning
   - *Lights, Coronagraph, Action! An Exoplanet Direct Imaging Demo*, September 2025, NASA's Universe of Learning
   - *Exoplanet Detectives: A Transit Method Activity*, September 2025, NASA's Universe of Learning
2. **David R. Coria**, N. Hejazi, I. J. M. Crossfield, M. Rhem; *The Wanderer: Charting WASP-77A b's Formation and Migration Using a System-Wide Inventory of Carbon and Oxygen Abundances*, 2024, ApJ, 974, 151
3. **David R. Coria**, I. Crossfield, J. Lothringer et al.; *The Missing Link: Testing Galactic Chemical Evolution Models with the First Multi-Isotopic Abundances in Dwarf Stars*, 2023 ApJ, 954, 121
4. **David R. Coria**, C. Bergmann, C. Tinney; *Veloce Quick Look App (2019),* Research Practicum, University of New South Wales, Sydney, Australia

Contributing Author Publications

Commented [DC9]: Remove doi; Add link to titles

5. Darío González Picos, Ian J.M. Crossfield, **David Coria** et al., 2026, The Isotopic and Elemental Abundances of Planet-Host Star TRAPPIST-1, Accepted to ApJ
6. Alex Polanski, Ian J. M. Crossfield, Andreas Seifahrt et al, **incl. D. Coria**, 2025, An Aligned Sub-Neptune Revealed with MAROON-X and a Tendency Toward Alignment for Small Planets, AJ, 170, 182
7. Neda Hejazi, Sébastien Lépine, Thomas Nordlander et al. **incl. D. Coria**, 2025, Abundance Measurements of the Metal-poor M Subdwarf LHS 174 Using High-resolution Optical Spectroscopy, AJ, 170, 18
8. Neda Hejazi, J. Xuan, **D. Coria**, et al, 2024, *Chemical Links between a Young M-type T Tauri star and its Substellar Companion: Spectral Analysis and C/O Measurement of DH Tau A*, ApJ, 978, 42
9. Neda Hejazi, Ian J. M. Crossfield, Diogo Souto et al. **incl. D. Coria**, 2024, *High-resolution Elemental Abundance Measurements of Cool JWST Planet Hosts Using AutoSpecFit: An Application to the Sub-Neptune K2-18b's Host M Dwarf*, ApJ, 973**,** 31

10. Neda Hejazi, I. Crossfield, T. Norlander, M. Mansfield, **D. Coria** et al.; *Detailed Elemental Abundances of a Super-Neptune Host Star Using High-Resolution, Near-Infrared Spectroscopy*, 2023, ApJ, 949, 79
11. Alex S. Polanski, Jack Lubin, Corey Beard, et al; *The TESS-Keck Survey. XX. 15 New TESS Planets and a Uniform RV Analysis of All Survey Targets*, ApJS, 272, 32
12. I. Crossfield, M. Malik, M. Hill, S. Kane, B. Foley, A. Polanski, **D. Coria**, et al.; GJ 1252b: A Hot Terrestrial Super-Earth with no Atmosphere, ApJL, 937, L17, 2022